APEX RETAIL GROUP | INDEPENDENT HOMEWORK ASSIGNMENT

DOCUMENT: CLASS HOMEWORK (SESSION 6)

SCENARIO: OMNI-CHANNEL RETAIL PERFORMANCE & CUSTOMER TIER SEGMENTATION

ASSIGNMENT OVERVIEW: You are assisting the Senior Business Analyst at Apex Retail Group. Leadership wants to understand how different customer tiers (Platinum, Gold, Silver, Bronze) interact with our product categories and which channels drive the highest order values.

You must load the clean transactional dataset, execute grouped aggregations, rank performance by customer segment, and prepare an analytical report for the Category Manager.

RAW TRANSACTION DATASET:

```python
import pandas as pd

omni_sales_data = {
    "order_id": ["ORD-801", "ORD-802", "ORD-803", "ORD-804", "ORD-805", "ORD-806", "ORD-807", "ORD-808", "ORD-809", "ORD-810", "ORD-811", "ORD-812"],
    "channel": ["Online", "In-Store", "Online", "In-Store", "Online", "Online", "In-Store", "In-Store", "Online", "In-Store", "Online", "In-Store"],
    "customer_tier": ["Gold", "Silver", "Platinum", "Bronze", "Gold", "Platinum", "Silver", "Bronze", "Gold", "Platinum", "Silver", "Bronze"],
    "department": ["Electronics", "Apparel", "Electronics", "Home", "Apparel", "Electronics", "Home", "Apparel", "Home", "Electronics", "Apparel", "Home"],
    "net_revenue": [1450.00, 180.00, 2100.00, 95.00, 220.00, 1850.00, 110.00, 140.00, 130.00, 3100.00, 195.00, 85.00],
    "discount_applied": [50.00, 10.00, 100.00, 5.00, 15.00, 75.00, 0.00, 10.00, 10.00, 150.00, 12.00, 5.00]
}
```

PROBLEM 1: CUSTOMER TIER SEGMENTATION & REVENUE SUMMATION

In a Jupyter Notebook named `homework_session6_yourname.ipynb`:

1. Load `omni_sales_data` into DataFrame `df_omni`.
2. Group `df_omni` by `customer_tier` and calculate:
    - Total net revenue generated (`net_revenue`: `sum`)
    - Average net revenue per order (`net_revenue`: `mean`)
    - Total order count (`order_id`: `count`)
3. Sort the resulting customer tier summary by total net revenue in descending order.

PROBLEM 2: MULTI-LEVEL CHANNEL & DEPARTMENT AGGREGATION

1. Group `df_omni` by both `channel` and `department` (multi-index grouping).
2. Calculate total `net_revenue` and average `discount_applied` for each channel-department pair.
3. Print the resulting multi-index summary DataFrame.

PROBLEM 3: STRATEGIC INSIGHT REPORTING

Write a 150-word formal analytical report for the Senior Business Analyst containing:

1. Identification of the most lucrative customer tier based on total revenue and average order value.
2. A comparison between Online versus In-Store channels across departmental revenue generation.
3. Recommendations on which customer tier and sales channel deserve increased marketing acquisition spend.

SUBMISSION DEADLINE: Prior to Session 7.